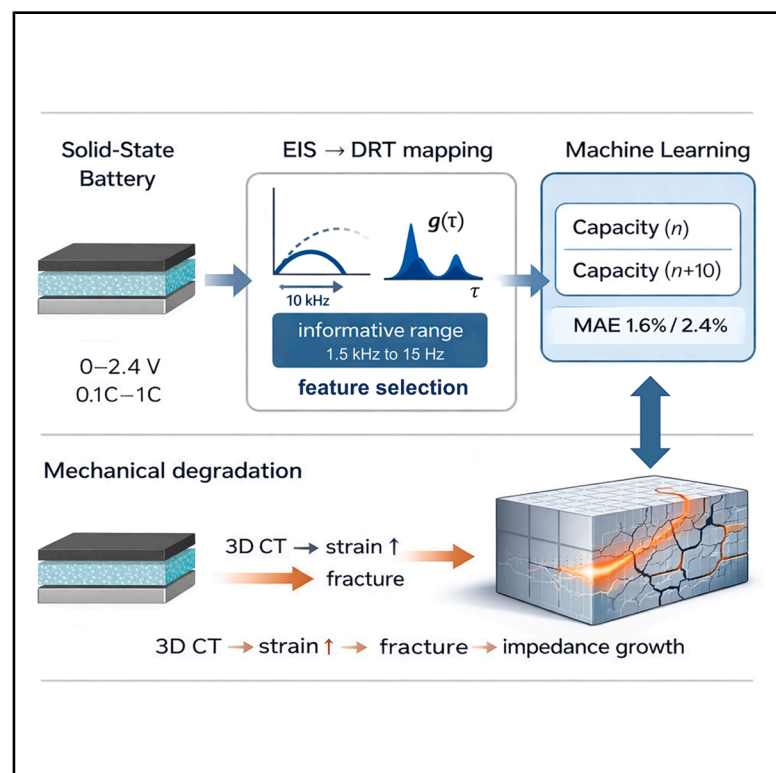


Predicting battery performance with small datasets by enhancing machine learning with distribution of relaxation times analysis

Graphical abstract



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In brief

Zhu et al. combine electrochemical impedance measurements with machine learning to predict the performance of solid-state batteries using only small datasets. By selecting physically meaningful features through relaxation-time analysis, the approach enables fast and accurate capacity diagnostics and forecasting, offering a practical pathway toward efficient battery testing and management.

Highlights

- Machine learning and electrochemical approaches are combined to predict battery capacity
- The model can predict cell capacity with an error of 2.4%
- Distribution of relaxation time analysis improves feature selection and training accuracy
- X-ray computed tomography reveals mechanical degradation and aids in interpreting the model



Article

Predicting battery performance with small datasets by enhancing machine learning with distribution of relaxation times analysis

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<https://doi.org/10.1016/j.xcrp.2026.103334>

SUMMARY

Accurate battery diagnostic models often struggle in capturing complex degradation mechanisms. Here, we report a combined approach using distribution of relaxation time (DRT) analysis and machine learning (ML) of electrochemical impedance spectroscopy (EIS) data to diagnose and predict the performance of solid-state batteries. The dataset has 112 impedance measurements collected over 100 cycles from eight TDK CeraCharge batteries. Using EIS features and the applied charge-discharge rate, the models estimate battery capacity at a given cycle and predict the capacity after the subsequent ten cycles. ML methods, including Bayesian regression, Gaussian process regression, neural networks, and decision trees, achieve accurate diagnostics (mean absolute error 1.6%) and prediction (2.4%). DRT-based feature selection identifies the most informative frequency range (15 Hz–1.5 kHz) and improves training efficiency. Tomography reveals mechanical degradation, i.e., volume expansion and fractures, which increases impedance and contributes to capacity loss. This approach enables accurate, data-driven battery diagnostics and forecasting.

INTRODUCTION

Solid-state batteries (SSBs) are widely considered a promising alternative to conventional lithium-ion batteries. Compared with batteries using organic liquid electrolytes, SSBs offer improved safety due to their solid electrolytes (SEs). Consequently, the stability of SEs against electrodes has become a major focus of recent research.^{1,2} However, understanding and predicting the performance degradation of SSBs remain challenging because their aging involves strongly coupled electrochemical, chemical, and mechanical processes.

Reliable diagnostic methods at the laboratory scale are therefore highly desirable. Fast and non-destructive diagnostics could accelerate the evaluation of new battery designs and complement the material-intensive and time-consuming experiments typically required in industrial development. Such tools are also critical for practical applications: defective cells must be identified during production, and cells with accelerated degradation should be detected and managed by battery management systems to ensure safe battery pack operation.

Traditional approaches to predicting battery performance rely mainly on physics-based models that describe degradation mechanisms such as interphase growth^{3,4} and lithium plating.^{5,6}

However, applying these models to SSBs is particularly difficult because mechanical degradation, including fracture and deformation, is strongly coupled with electrochemical and chemical processes. This complexity makes accurate physics-based modeling of SSB degradation challenging.⁷

Recent advances in computing have enabled data-driven approaches that show strong potential for battery diagnostics and lifetime prediction. Zuo et al.⁸ reviewed applications of text-mining-based large language models in battery research, highlighting their ability to integrate computational and experimental tools and accelerate materials' discovery. Severson et al.⁹ demonstrated that early-cycle data can predict battery lifetime using machine learning (ML). Gasper et al.¹⁰ showed that impedance-based features enable accurate capacity prediction with a mean absolute error of 1.9%, while Zhu et al.¹¹ estimated battery capacity from voltage relaxation curves with errors below 1.7%. These studies demonstrate the promise of ML for battery diagnostics, particularly when physically meaningful features are incorporated to improve interpretability.¹²

Despite this progress, most data-driven studies focus on conventional lithium-ion batteries, while diagnostic frameworks for SSBs remain limited. In particular, methods that combine physically interpretable electrochemical features with ML to diagnose



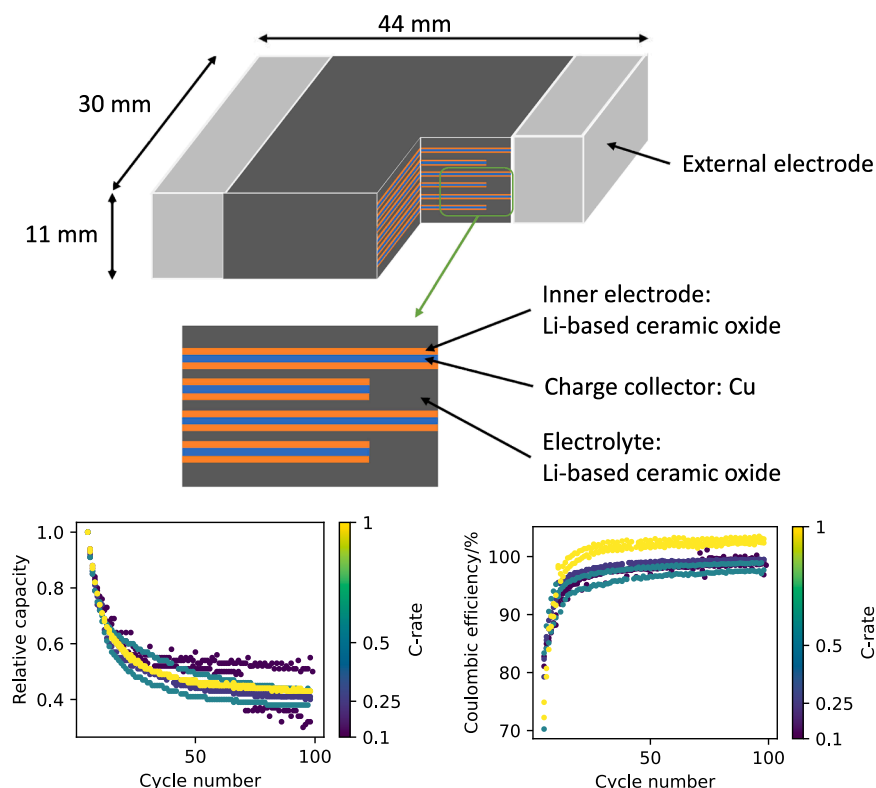


Figure 1. Cell concept and performance for increasing C-rates

Top: cell structure of the TDK CeraCharge solid-state battery, adapted and redrawn from Tan et al.² Lower left: discharged relative capacity vs. cycle number. Lower right: coulombic efficiency vs. cycle number.

and predict SSB performance are still underexplored.¹³ Developing such approaches could improve both prediction accuracy and understanding of the degradation processes in the SSB systems.

Here, we report a data-driven diagnostic framework that combines electrochemical impedance spectroscopy (EIS), distribution of relaxation time (DRT) analysis, and ML to diagnose and forecast the performance of SSBs. We study eight commercial CeraCharge SSBs from TDK,¹⁴ of which feature a multilayer symmetric architecture with a ceramic-based oxide SE. The cells are cycled for 100 charge-discharge cycles, and impedance spectra are collected throughout cycling. Features derived from the impedance spectra are used to diagnose the current capacity and predict the capacity ten cycles ahead. DRT analysis identifies the most informative frequency range (approximately 15 Hz–1.5 kHz), enabling efficient feature generation and improved model performance. The resulting models achieve a mean absolute error of 1.6% for capacity diagnostics and 2.4% for capacity prediction. X-ray computed tomography (XCT) further reveals volume expansion and mechanical fracture during cycling, which contribute to impedance growth and capacity loss. Together, these results demonstrate that integrating impedance-based electrochemical analysis with ML provides an effective framework for diagnosing degradation and predicting the performance of SSBs.

RESULTS AND DISCUSSION

Data generation and preprocessing

Ling¹⁵ pointed out the challenge of data scarcity in battery informatics. Despite the recent advancements, one of the major

obstacles is the limited availability of *comprehensive* and *high-quality* battery data. Vargas-Barbosa¹⁶ emphasized another problem—the variety of benchmarks and assessment methods across battery techniques reduces the comparability of SSB data and equitable comparisons. As there are only a few SSBs on the market, which vary extensively in terms of material choice and cell concepts, standardized test protocols and evaluation categories are not yet available for them. In addition to the choice of measuring procedure, the quality of test samples also influences data quality. Given the challenges in consistently controlling the quality of lab-produced SSBs, we opted to conduct our experiments using a commercially available SSB for small consumer electronic boards, the TDK CeraCharge.¹⁴ This cell has a multilayer and symmetrical structure with an oxide ceramic-based SE, as shown in Figure 1. The cell is mounted on a printed circuit board (PCB) board for testing, and the sample configuration is shown in Figure S1. This cell has a stable discharge capacity and a long cycle life, while using the suggested specifications of the manufacturer (potential window of 0–1.6 V), designed for more stable and long-life applications.

For our purposes, it is more meaningful and efficient to test the cell using its *full potential range*, i.e., extended beyond the suggested specifications, as the goal of this work is to develop a characterization and diagnosis method, covering all possible degradations that could be expected to be seen during extended cycling. An explanation of our specific SSB cycling protocol is found in the [cell study and analysis](#) section. We cycled the cells using the constant current/constant voltage (CCCV) protocol for both charging and discharging processes within 0–2.4 V, where the chosen C-rates were 0.1, 0.25, 0.5, and 1 C. After the cell was

fully charged or discharged, we applied EIS measurements after a sufficient relaxation period. Figure S2 shows one charge-discharge cycle of a cell. In the beginning phase of the cycling (the first 10 cycles), we performed EIS measurements after every charge-discharge process, during which the cells continue their formation. (Due to their solid-state character, the cells have an extended formation period at the beginning phase of usage compared to cells with liquid electrolytes, while the SE gets activated slowly.) Thus, the cells' impedance changes more in the initial phase, as the conductivity of the cells changes. In the following phase of the cycling, we performed EIS measurement every 10 cycles to save the testing time, as the test bench used can only conduct one EIS measurement at the same time. Figure S6 shows the entire test procedure.

We didn't apply the potential window suggested by the manufacturer; instead, we wanted to design an efficient test protocol and explore the full working range of the cells. Therefore, we investigated them in the potential window of 0–2.4 V and compared their performance with information from the corresponding patent and previous reports.^{17–20} To characterize cell degradation under different operating stresses, we designed the experiments with the C-rate as the controlling variable. The chosen rates were 0.1, 0.25, 0.5, and 1 C. Each C-rate experiment was performed in duplicate. Thus eight TDK CeraCharge cells were cycled with varied charging and discharging C-rates, after forming, in a temperature- and humidity-controlled environmental chamber (25°C and 30% humidity). The cells' cycling performance is shown in Figure 1. For a deeper insight into cell degradation, EIS was conducted periodically during cycling. Figure S4 shows that the cell impedance is highly dependent on the state of charge, especially in the smaller potential window of 0–1.6 V. With more cell capacity and higher cell voltage, the impedance gets lower. Besides, the cell voltage gets unstable during the EIS measurement, which makes the EIS data unreliable. Figure S5 compares the cells' impedance after the cell is charged to different capacities and has undergone different times of relaxation. The fully discharged cell has the most stable impedance. Therefore, we acquired the EIS data when the cell had been fully discharged and had undergone a 15 min waiting time to ensure the comparability of data and the reliability of results. The rated capacity was determined individually for each cell, based on its discharged capacity after formation.

We adopted a constant current cycling protocol with an expanded operating voltage range (as compared to the suggestion of 1.6 V from the manufacturer). Additional features of the charging process are to be expected above 1.5 V. An example is given in Figure S3, which shows 10 charge-discharge cycles of one cell (voltage vs. capacity) during operation. The cell voltage rises up to 1.5 V with a very low charge capacity and has, afterward, a relatively linear and slow voltage increase while charging with constant current. This voltage increase is more pronounced at higher cycle numbers, and thus, less capacity could be accessed at the same voltage level after cycling. Both the charged and discharged capacities decrease upon aging. We therefore assume that the potential-dependent capacity fade follows the same trend as described by Kobayashi et al.¹⁸ and Sørensen et al.¹⁹ for the electrode active material $\text{Li}_3\text{V}_2(\text{PO}_4)_3$.

Both report that the phase evolution during cycling follows the two-phase transition for the extraction of the first two lithium ions, while a solid solution behavior was discovered during the extraction of the last lithium ion in the $\text{Li}_3\text{V}_2(\text{PO}_4)_3$ cathode (when the cell potential is above ca. 2.0 V). The phase transition is fully reversible for the extraction of first and second lithium ions, while the solid-solution behavior observed during the extraction of the third lithium ion leads to irreversible structural hysteresis between charge and discharge.¹⁹ The volume of the cell changes continuously when transitioning between $\text{V}_2(\text{PO}_4)_3$ and $\text{Li}_3\text{V}_2(\text{PO}_4)_3$ during lithium reinsertion. This transition leads to a change in the electrode crystal structure and an irreversible capacity loss for the cell, and thus the loss of lithium inventory. We intentionally cycled the cells up to 2.4 V,¹⁸ where irreversible lithiation happens. Thus, irreversible capacity loss is expected in all our tested cells. Figure 1 (lower left and lower right) show the capacity fade and the evolution of their coulombic efficiency over 100 cycles. The severe capacity loss in the early cycles should be caused by the irreversible loss associated with the insertion and extraction of the third lithium ion.

Besides analyzing the battery charging and discharging processes during aging, we also characterized their impedance development. EIS was performed in galvanostatic mode (GEIS, single-sine) with an AC current amplitude of 5 μA . Spectra were collected from 10 kHz to 10 mHz using logarithmic spacing with 6 points per decade (total 6 decades, 36 points). At each frequency, 5 acquisitions/averages ($N_a = 5$) were applied by the instrument. The impedance rises in different patterns in the low-, middle-, or high-frequency ranges (Figure 2 [top]), which reflects different reaction mechanisms in the battery. The impedance development of cells operated under different C-rates is given in Figures S8 and S9. Figure 2 (top) shows the impedance of a cell cycled at 0.1 C: the impedance measured at a certain frequency rises almost linearly during aging, for example, the impedance measured at 130 Hz. To ensure the reliability of the impedance data, we used the Kramers-Kronig (KK) test to prove the linearity, stability, and causality of the tested systems. The KK test is a mathematical consistency check used to verify whether a set of frequency-dependent impedance data obeys the fundamental laws of causality and linearity. Schönleber et al. have developed a tool called "Lin KK" to apply the KK principle in validating the quality of impedance data.²¹ This tool is available in various environments, such as MATLAB or Python package "impedance.py."²² LinKK evaluates the KK consistency of EIS data by fitting a linear KK-consistent model to the measurements and analyzing the residuals. If the residuals are small, the data are consistent with the KK relations—meaning they come from a causal, linear, stable system. With this help, we observe that the lower frequency range data are less reliable according to the KK test, particularly evident for cells cycled at 1 C (Figures S10 and S11). For example, for the cells cycled at 1 C (Figure S11C), the residuals between the measured and calculated impedance parts by the KK method reach almost 1% at 0.01 Hz, while the residuals lie around –0.1% to 0.2% in the medium frequency range. Although EIS spectra were measured down to 10 mHz, the impedance data below approximately 0.2 Hz exhibited significantly reduced reliability according to the KK consistency test (Figures S10 and S11). The increased

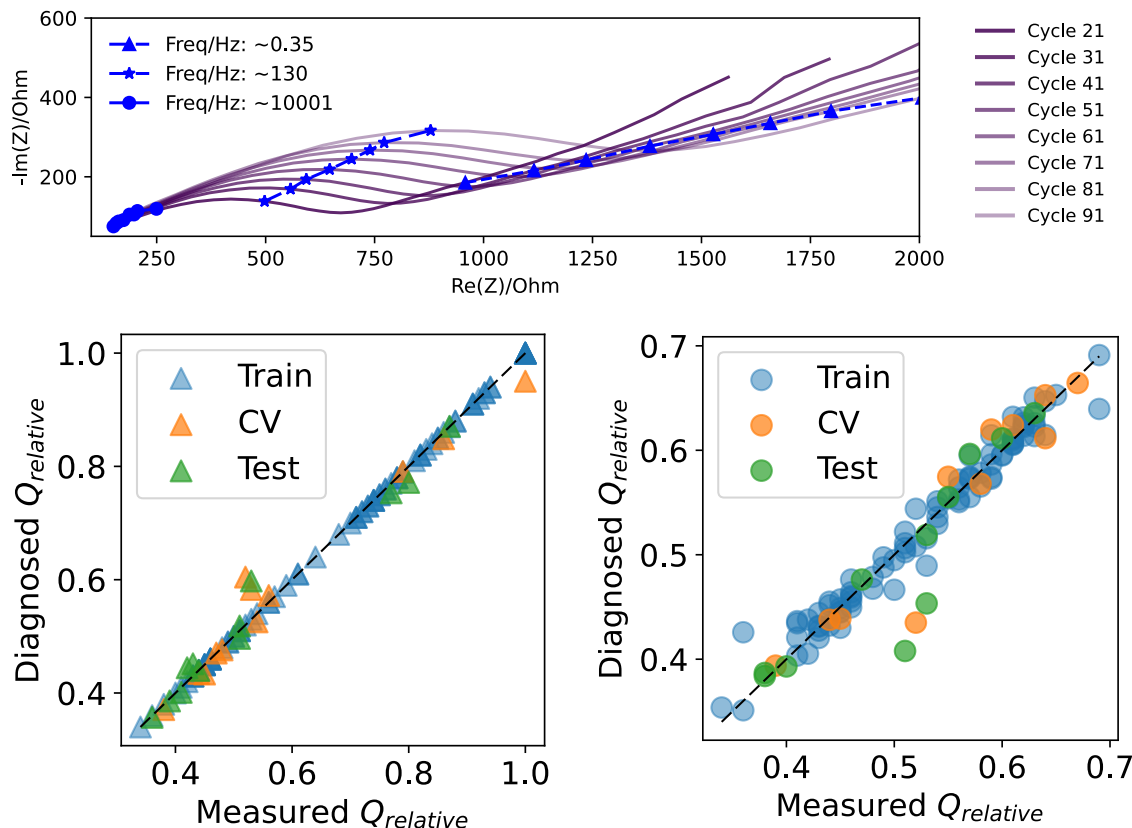


Figure 2. Representative EIS data and modeling results

Top: impedance spectra of a representative SSB cell, EIS measured at 0% SOC, cell cycled with 0.1 C, with marked impedance data points in the low-, middle-, and high-frequency range over a number of different cycles. Lower left: measured and diagnosed relative discharged capacity using Gaussian process regression (GPR), with DRT instructed feature engineering, data divided into training, cross validation (CV), and test sets. Lower right: measured and predicted relative discharged capacity after next 10 cycles using Bayesian regression (BR), with DRT instructed feature engineering, data divided into training, CV, and test sets.

KK residuals in this frequency range indicate violations of the fundamental EIS assumptions of linearity, stability, and time invariance. These deviations are attributed to the long acquisition times at low frequencies, during which slow relaxation and diffusion-related processes can evolve, rendering the system effectively non-stationary. Consequently, impedance data below 0.2 Hz were excluded from subsequent analysis to ensure data quality and EIS validity. In total, we generated 112 impedance datasets for further analysis, which reflect the cells' performance after formation until the cells' 100th cycle life. The input factors of each dataset are the operating C-rate, the measured imaginary, and real impedance data points.

Feature engineering and ML modeling

We tested eight cells for 100 cycles in duplicate at four different C-rates (0.1, 0.25, 0.5, and 1 C). Impedance data collected after the formation process were selected for modeling, resulting in a total of 112 datasets gathered (for further cycling details, see methods: [cell study and analysis](#)). It should be noted that one cell exhibits a higher initial impedance than the other cells (see [Figure S8D](#)), likely due to manufacturing-related imperfections. This cell was retained in the analysis to evaluate the robustness

of the proposed modeling approach, which remains effective despite the presence of such outliers. The datasets are divided into training data, cross validation data, and test data groups (ratio 8:1:1). The ML target is to diagnose the relative discharged capacity of the cells based on their impedance data and operational C-rate. For the prediction task, the model inputs consist exclusively of impedance-derived features measured at cycle n and the corresponding operational C-rate. The target variable is the relative discharged capacity at cycle $n + 10$. The measured capacity at cycle n is not included as an input feature to the model. We observe that the cell impedance measured at a certain frequency increases in a relatively linear pattern as the cycle number rises ([Figure 2](#) [top]). Given the small dataset and the observed linearity, we hypothesize that a linear regression model would be effective for evaluating the relative capacity from EIS data. Additionally, we explore alternative modeling approaches, such as Bayesian regression (BR), Gaussian process regression (GPR), neural network (NN), and decision tree (DT).

For constructing models, we employ a feature-based methodology. As the impedance data below 0.2 Hz violates the fundamental EIS requirements of linearity, stability, and causality due to long acquisition times, as confirmed by the KK test

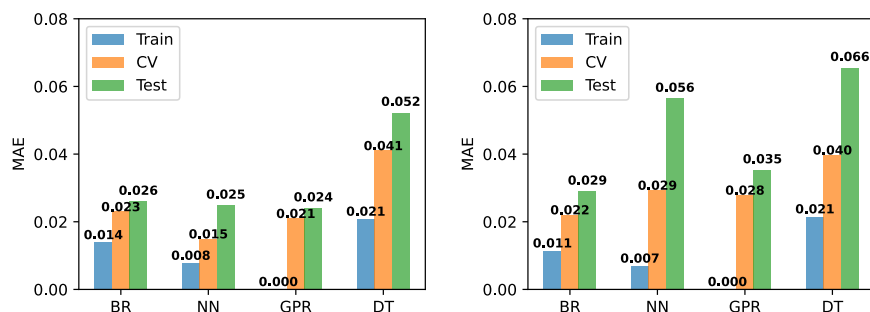


Figure 3. ML model results without feature reduction

Mean absolute error (MAE) of training, cross validation (CV), and test datasets in Bayesian regression (BR), neural network (NN), Gaussian process regression (GPR), and decision tree (DT) for SSB data obtained at 0.1 to 1 C, up to 100 cycles for diagnostic models (left) and prediction models (right).

(Figures S10 and S11), only impedance features above 0.2 Hz were used for ML model training. Our input features are the chosen impedance data points and the operational C-rate during the cycling process. The features, which represent both linear and nonlinear transformations of the raw data, are generated and integrated within a regularized linear framework. The feature engineering processes are designed within the pipeline of each model.²³ The feature polynomialization, regularization, and feature selection are included in data processing based on the model type. Figure S7 shows the data processing for the BR model with preprocessing and model-based data processing parts. The feature selection designed in the pipeline is to reduce the number of features but to retain the significant features for prediction from a statistical perspective. We want to emphasize that the distribution of the input data can significantly influence the learning process of both BR and GPR. Although most models do not strictly require the input data to follow a normal distribution, we observe that the shape of the data distribution can affect prediction accuracy. If the data are highly skewed or contain outliers, while BR and GPR still assume that the noise follows a normal distribution, the models may struggle to fit the data effectively. Additionally, the data distribution can influence the choice of kernel functions in GPR, making it important to preprocess and wrangle the data before applying any ML method. Therefore, only the regularization of all features is not enough. We apply the logarithm transformation of the raw impedance data to make sure the data obey a relatively normal distribution, as Figure S12 shows, which ensures the feasibility and reliability of BR and GPR modeling approaches.

Performance without feature reduction

As the impedance data show less linearity, stability, and causality below 0.2 Hz based on the KK test (Figures S10 and S11), we first adopted all the features above 0.2 Hz for training the ML models. Surprisingly, all chosen models perform very well for diagnosing the capacity. An overview of all model performance is given in Figure S13. We use the mean squared error (MSE) and the mean absolute error (MAE) of the test datasets to evaluate the models' performance (based on Equations 1 and S2 in supplementary materials). The evaluation results can be found in Table S1. The data are distributed in training set, cross validation set, and test set in a ratio of 8:1:1. The training dataset is used to teach the model by optimizing its parameters, while the cross-validation set helps tune hyperparameters and prevents overfitting by assessing the model's performance during training. Finally, the test set evaluates the model's performance

on unseen data, providing an unbiased estimate of its real-world effectiveness. In our work, the BR, NN, and GPR models have promising diagnoses (Figure 3 [left]), especially the GPR, which has only 0.1% MSE and 2.4% MAE on the test dataset, which is the error between diagnosed relative capacity and measured capacity of a cell from unobserved EIS data (Equations S1 and S2). These results are in line with the recent ML model using EIS data from Gasper et al.,¹⁰ which achieved accurate capacity prediction with an MAE of 1.9% by using an ensemble model. The BR model has an R-squared value of up to 0.96, which means that 96% of the variance in the target variable is explained by the model, reflecting a good fit to the data. This high value suggests that the approach has effectively captured the underlying patterns while incorporating uncertainty in the predictions. The DT model, despite its relatively worse performance, has the fastest calculation among these methods (Table S5)—a clear advantage when using larger datasets.

To predict the relative discharge capacity after the next 10 cycles, the BR approach offers significant advantages with an MAE of 2.9% (Figure 3 [right]), by providing complete probability distributions for parameters, flexible integration of prior knowledge, and handling uncertainty effectively. It provides posterior probability distributions for parameters rather than just point estimates. This relationship means that the model can offer more comprehensive information, including a degree of uncertainty in parameter values. Another important reason is that the BR model incorporates an additional feature selection step in its pipeline, allowing it to statistically select the most important features based on the coefficient weights. This selection enables the model to compute faster while avoiding fitting distortion.

Performance with DRT-instructed feature selection

Although the previous section demonstrated high diagnostic and predictive accuracy, processing data that spans a broad frequency domain significantly increases the computational complexity and prolongs model training time. Likewise, conducting EIS tests across the entire frequency domain is time-consuming, and using the full dataset is computationally expensive, especially in industrial scenarios. Therefore, we intend to reduce the features for model training by reducing the data points in the raw datasets, i.e., select only a certain range of frequency data for modeling.

Figure 2 (top) shows the impedance development of a representative cell upon cycling. The marked impedance values, measured at low, middle, and high frequency (0.3, 130, and 10 kHz), increase linearly during cell aging, albeit with different

slopes. Similar to liquid electrolyte batteries, the impedance developments of oxide-based SSBs at different frequency ranges are indicative of different reactions and mechanisms. The SE aging can be observed in the high-frequency range, i.e., the increase in grain boundary resistance and the increase of the bulk resistance results in the shift of the whole impedance spectra along the x axis.²⁴ Aging due to phase degradation, moisture absorption, or lithium depletion can increase the bulk resistance of the electrolyte. Considering the brittle nature of the SE, structural change factor in as a cause for higher grain boundary resistance. All aforementioned factors would manifest as an increase in the high-frequency resistance of the Nyquist plot, reflecting the deteriorating ionic conductivity of the SE. Regarding the middle frequency impedance growth, increased charge transfer resistance is seen as the main reason.²⁴ It is caused by multiple interfacial degradation phenomena, e.g., the formation of interfacial resistance layers between the SE and electrodes. Besides, the phase transition in the cathode material for the insertion of the second lithium ion can also lead to irreversible structural changes and hinder the lithium ion transport.¹⁹ The low-frequency impedance is well known to be related to diffusion processes. Warburg impedance, which appears as a linear tail in the low-frequency region, indicates the difficulty in lithium ion diffusion through the solid-state materials. Over time, the diffusion of lithium ions, through the SE or within the electrode materials, can become impeded due to structural changes, e.g., void formation or pore clogging. Of course, the degradation mechanisms are coupled and cannot be purely divided based on frequency, as other aging mechanisms, such as the loss of active material may result in increased amplitudes in the overall EIS spectrum. However, the various trends in impedance changes in different frequency ranges reflect the different contributions of the various aging mechanisms to the overall degradation of the cell. For this reason, we have adopted statistical analysis and DRT analysis to find out how the impedance at different frequencies develop. Maradesa et al.²⁵ reviewed the DRT methods for analyzing EIS data, a non-parametric approach that maps impedance spectra into time-constant distributions to reveal the underlying electrochemical processes. Compared to equivalent circuit models, DRT offers simpler interpretation, as it does not need the definition of an equivalent circuit model and scalability to large datasets.

To further reduce the number of features, we apply DRT analysis, which offers the opportunity to select features that are interpretable. The DRT method transforms impedance data from the frequency domain to the time-constant domain, as explained in the section on DRT. By representing the impedance response as a distribution of relaxation times, DRT helps separate partially overlapping features along the timescale. This allows a more detailed interpretation of electrochemical processes and aging mechanisms without assuming a direct one-to-one link between impedance features and specific physical processes.

In a DRT plot, the distribution function $g(\tau)$ represents the distribution of resistive contributions as a function of the relaxation time τ . A DRT peak represents the superposition of a large number of elementary relaxation contributions associated with the same electrochemical process. However, contributions from different electrochemical processes may overlap within the

same time-constant range and therefore appear as a single DRT peak. Moreover, the resistance associated with a DRT feature is given by the area under the peak rather than its peak height, in accordance with the mathematical definition of the DRT formalism.

Within the framework of DRT analysis, a distinction must be made between Debye and non-Debye processes. Debye-type processes are characterized by a single relaxation time and can be adequately modeled by a resistance in parallel with a capacitor ($R||C$), giving rise to symmetric and well-centered peaks in the DRT spectra. In practical battery systems, many interfacial processes deviate from ideal Debye behavior and are better described by a resistance in parallel with a constant phase element ($R||Q$), reflecting a distribution of relaxation times due to, for example, electrode heterogeneity or microstructural effects.

Mass-transport phenomena such as Warburg-type diffusion exhibit a fundamentally non-Debye behavior. Unlike interfacial charge-transfer or surface-film processes, Warburg diffusion does not appear as a symmetric peak in the DRT spectrum. Instead, it typically manifests as a broad low-frequency contribution or tail, reflecting semi-infinite or finite-length diffusion processes in the electrochemical system.^{26–28}

Several studies have shown that diffusion-related impedance contributions can influence the DRT inversion, particularly near the low-frequency boundary, depending on the regularization strategy and frequency window employed.²⁹ As a consequence, Warburg-type elements are sometimes mathematically removed or treated separately prior to DRT analysis to avoid distortion of higher-frequency features.^{30,31} Nevertheless, it has also been reported that in most practical cases, the impact of low-frequency diffusion on intermediate-frequency DRT peaks is limited, provided that the frequency range is sufficiently broad and the inversion is properly regularized.²⁹

In the present study, employing symmetric cells with a minimum frequency of 10 mHz, diffusion-related impedance contributions may affect the inversion outcome at long relaxation times. However, consistent with the literature, their influence on the intermediate relaxation-time range considered in this work is expected to be limited. The EIS was measured after a sufficient relaxation period (see Figure S5) following cell discharge to mitigate the influence of diffusion-related processes on the EIS results, in line with the selection of this region for feature extraction and subsequent data-driven modeling.

In the present study, DRT spectra were calculated using the Gold algorithm.³¹ Figure 4 compares the DRT spectra of two symmetric cells operated at 0.1 and 0.25 C. The distribution function $g(\tau)$ increases progressively with cycling. In particular, in the relaxation-time range around $\tau \approx 10^{-3}$ s, the integrated contribution in this area increases from cycles 21 to 91, indicating the gradual development of resistive processes within this time-constant range rather than the emergence of a single isolated process. Similarly, Figure S15 shows that for cells operated at different C-rates, $g(\tau)$ increases with cycling between $\tau = 10^{-4}$ s and 10^{-2} s.

Based on these observations, the impedance features corresponding to relaxation times between 10^{-4} and 10^{-2} s were selected for feature extraction. This relaxation-time range is influenced by a broader frequency spectrum but is

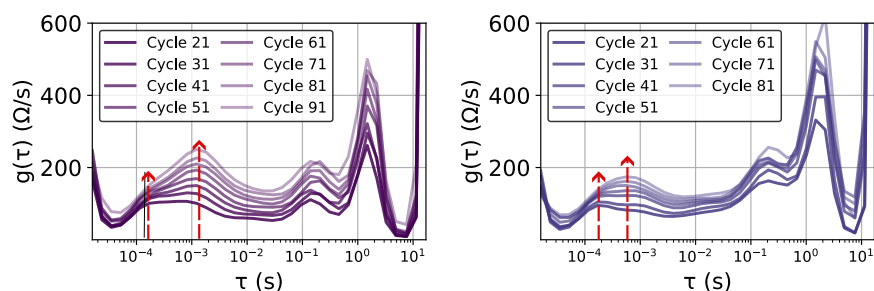


Figure 4. DRT analysis: the distribution function vs. time constants

Shown at regular intervals during the cycling of the chosen cells, cycled at two example C-rates. Left: 0.1 C; right: 0.25 C.

predominantly associated with frequencies between approximately 1.5 kHz and 15 Hz ($2\pi f\tau = 1$), where interfacial and charge-transfer-related processes are expected to dominate. Moreover, this frequency window exhibits higher reliability according to the KK consistency test (Figures S10 and S11), supporting its suitability for reduced-order model training.

With DRT-instructed feature selection, the training speed has been improved enormously for the BR and GPR models. For example, the training of the GPR model has been accelerated by 80% on a computer with Intel(R) core (TM) i7 CPU, Intel(R) UHD Graphics630 GPU, and RAM 64 GB. A comparison of the model training times, before and after feature reduction, is given in Table S5. Compared to the training outcomes obtained without DRT feature selection, the BR and GPR models demonstrate improved diagnostic performance, whereas the NN model does not, as illustrated in Table S3. The GPR model achieves a precise diagnosis of cell capacity based on unobserved impedance, exhibiting a remarkably low MSE of 0.06% and an MAE of 1.6%, as shown in Figure 5. The capacity prediction in the next 10 cycles has only been slightly improved in the BR model, resulting in an MSE of 0.16% and an MAE of 2.4%.

Based on the training with various models and different feature selections, we conclude that when using full-frequency impedance data, all models predict the capacity accurately and demonstrate excellent fitting. Through DRT-based feature analysis, training speed can be significantly accelerated for the BR and

GPR models, and the influence of unreliable data from certain frequency domains can be mitigated. Next, it allows better access to the interpretation of the cell aging than without. In this case, the selected frequency range (1.5 kHz–15 Hz) impedance possesses clearer trends during cell aging and higher data quality, considering the KK test of system linearity, stability, and causality (Figures S10 and S11). Despite the small datasets, the best model for capacity diagnoses, the GPR approach, achieves an MSE of 0.06% and an MAE of 1.6% on the test datasets (Figure 2 [lower left]). For predicting the capacity after the next 10 cycles, the BR model exhibits notable advantages with or without DRT feature selection, emerging as the best model capable of delivering dependable and precise predictions. The best case delivers an MSE of 0.16% and an MAE of 2.4% in predicting the cell capacity from unobserved impedance data (Figure 2 [lower right]).

Ex situ investigation and degradation analysis

Many ML approaches are considered “black boxes,” making it difficult to interpret the predicted results. On the other side, for some critical applications, such as battery management in electric vehicles, explainability and interpretability are crucial for ensuring the reliability and safety of the system. Therefore, understanding the underlying physical mechanisms and connecting them with the model training results is essential for all ML work. The degradation mechanisms in SSBs can be described as mechanical failures, interfacial resistance buildup, electrolyte

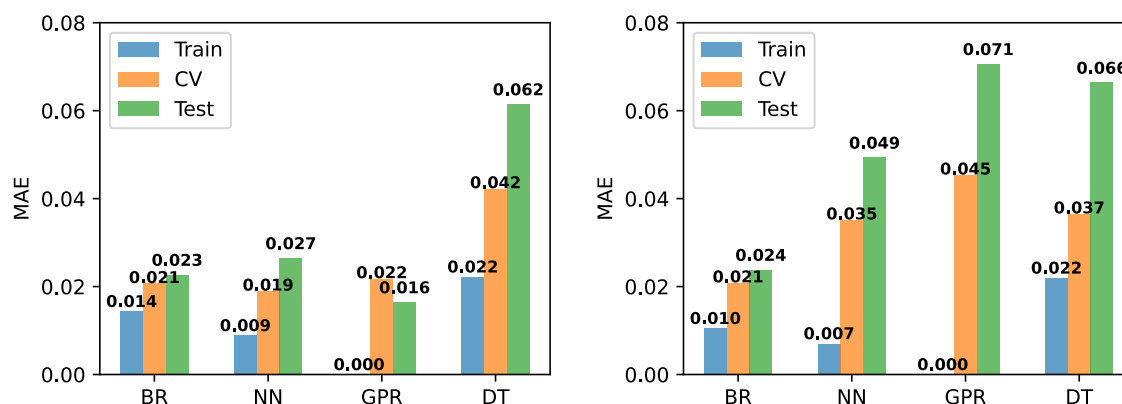


Figure 5. ML model results with feature reduction based on DRT analysis

MAE of training, CV, and test datasets in BR, NN, GPR, and DT for SSB data obtained at 0.1 to 1 C, up to 100 cycles for diagnostic models (left) and prediction models (right).

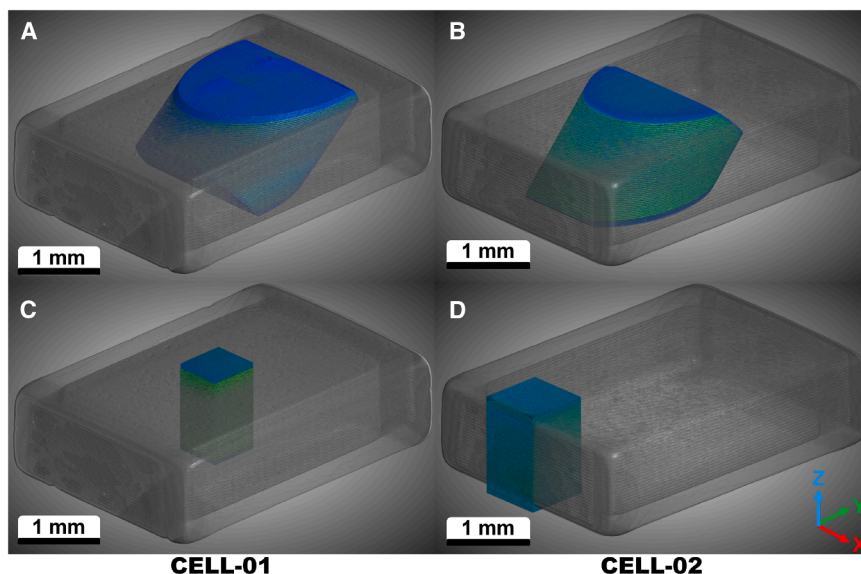


Figure 6. 3D rendered images of two TDK CeraCharge SSB from CT measurements

Both cells were cycled at 0.1 C, while the impedance of cell-01 is shown in Figure 2 (top). The LCT machine produced grayscale images with 4.8 μm resolution, and the blue-green zones were produced with 1 μm resolution and then used for further evaluation. The blue-green zone of (D) has a resolution of 0.72 μm taken from the SCT dataset. (A) and (B) are used for qualitative evaluation, whereas (C) and (D) are used for quantitative evaluation.

decomposition, cathode material degradation, loss of lithium inventory, etc. These mechanisms are related to cell structure and materials. The degradation phenomena are coupled and together lead to increased internal resistance and capacity fading.

Considering the multilayer ceramic oxide-based cell concept, as shown in Figure 1 (top), it was impossible to open the cell non-destructively to do a postmortem analysis after cell degradation. We also tried to analyze the internal structural changes of a cell by applying a focused ion beam (FIB) through the solid-electrolyte layer and using a scanning electron microscope (SEM) to investigate the cross-sectional surface morphology. However, the material proved to be too soft, and its structure was destroyed by the FIB. Therefore, we adopted the XCT technique to study the effect of cycling on SSBs.

The XCT measurements used two different machines: the commercial ZEISS Xradia 620 Versa X-ray microscope and synchrotron XCT at BAMline20.³² The latter will be referred to as synchrotron computed tomography (SCT), and the other is the laboratory computed tomography (LCT) in this study. The SCT provides a parallel monochromatic X-ray beam with energies ranging from 8 to 60 keV, with an energy resolution $\Delta E/E$ of around 4%, thanks to a W/Si double multilayer monochromator. The 3D reconstructions of the SCT dataset were performed using an in-house software coded in Python, built upon the “Tomopy” package. The LCT machine is equipped with an X-ray source from Nordson Dage “QuadraNT” 4th gen with a running voltage between 30 and 160 kV, where the maximum power is 25 W. It also has a built-in filter disk consisting of 12 different filters to alter the transmitted photon energy, thus producing a better image quality. Additionally, there are two ways to obtain the radiographs. There was a flat-panel detector (FPX) and a charge-coupled device camera, where the latter captured the visible light that had been converted from X-rays using a scintillator, located in the multiple magnification objectives (0.4X, 4X, 20X,

and 40X). A zoom-in scan method was utilized in this study using both the FPX and 4X, as depicted in Figures 6A and 6B by the grayscale and blue-green colored images, respectively. This procedure implies that two volume images can be overlapped onto each other, where one has a smaller field of view (FoV) and a smaller voxel size at the precise location, based on the larger FoV scan, without removing the sample from the machine. The 3D image reconstruction is performed automatically using the built-in algorithm by Zeiss.

There are three types of images in total: two types were taken with the LCT, and the third comes from the SCT. The main difference lies in their voxel size, which is 4.8 μm and 1 μm from the LCT scans, whereas the SCT scan yielded a voxel size of 0.72 μm . Details of the X-ray scanning parameter can be seen in Table S6 (supplemental information). The LCT scans were performed both on pristine and cycled SSBs, where unfortunately, the damage on cell-02 appeared at the other location than the zoomed-in images. Therefore, the SCT scan was performed on the cell-02 around the damaged region to be able to quantify the observed damage. In this way, the difference in image clarity between LCT and SCT datasets can also be further evaluated. All the image processing tasks were carried out using the Dragonfly 3D World software, version 2024.1,³³ and a digital volume correlation (DVC) module named SPAM^{34,35} was used to quantify the effect of battery cycling.

Qualitative evaluation

Figure 7 displays the difference between the pristine and cycled conditions of the cell-01 battery. The corresponding slices were taken from the planes depicted by the red, blue, and green colored planes. Globally, many cracks appeared, both in- and out-of-plane view, after the cell was cycled. The dotted orange lines in Figure 7 highlight the area with potential weak points that may lead to mechanical degradation, e.g., due to crack formation. Several dark spots could be seen as voids filled with non-solid material, presumably gas. Additionally, the anode

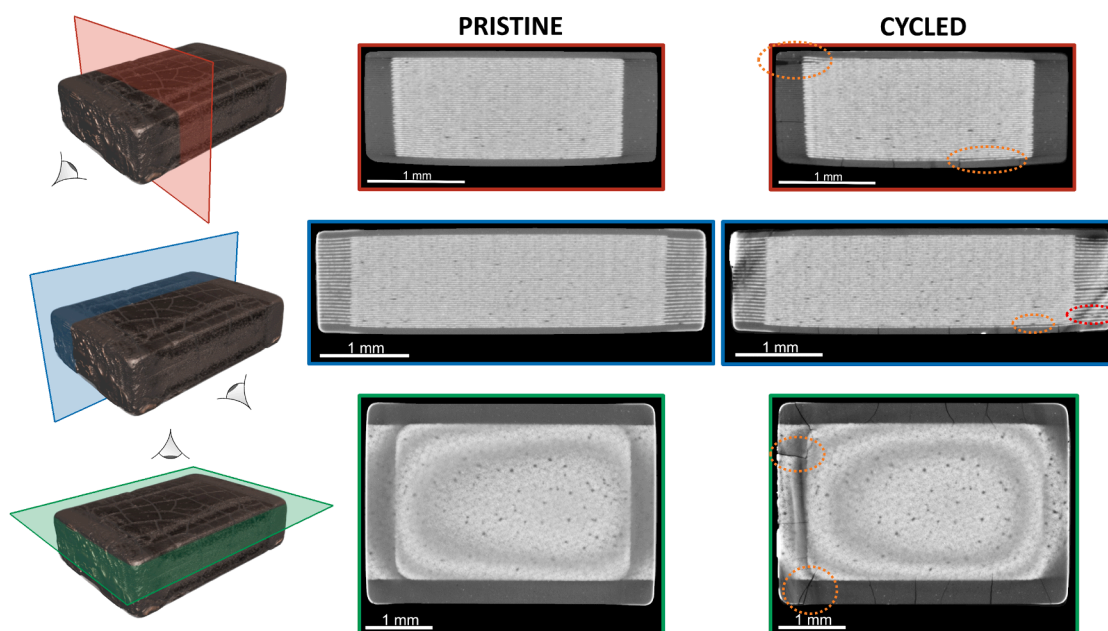


Figure 7. CT images of a pristine and a cycled cell-01

The cell was scanned at a fully discharged state at the corresponding slices with $4.8\ \mu\text{m}$ voxel size. The cycle process was carried out under 0.1 C and 100 cycles. The red (XZ plane), blue (YZ plane), and green (XY plane) slices are displayed on the top, middle, and bottom regions.

was found to become corrugated with cycling that can be seen inside the red dotted line.

In Figure 8, some potential weak points can be observed, inside the areas highlighted by the orange dotted lines, where mechanical cracks are also visible. The dark spots appear next to

the visible cracks, which look to be fractures in the SE and the electrode materials in both horizontal and vertical directions. The most significant damage can be observed inside the area with yellow dotted lines, in which a large expansion emerges after cycling. This observation is presumably linked to

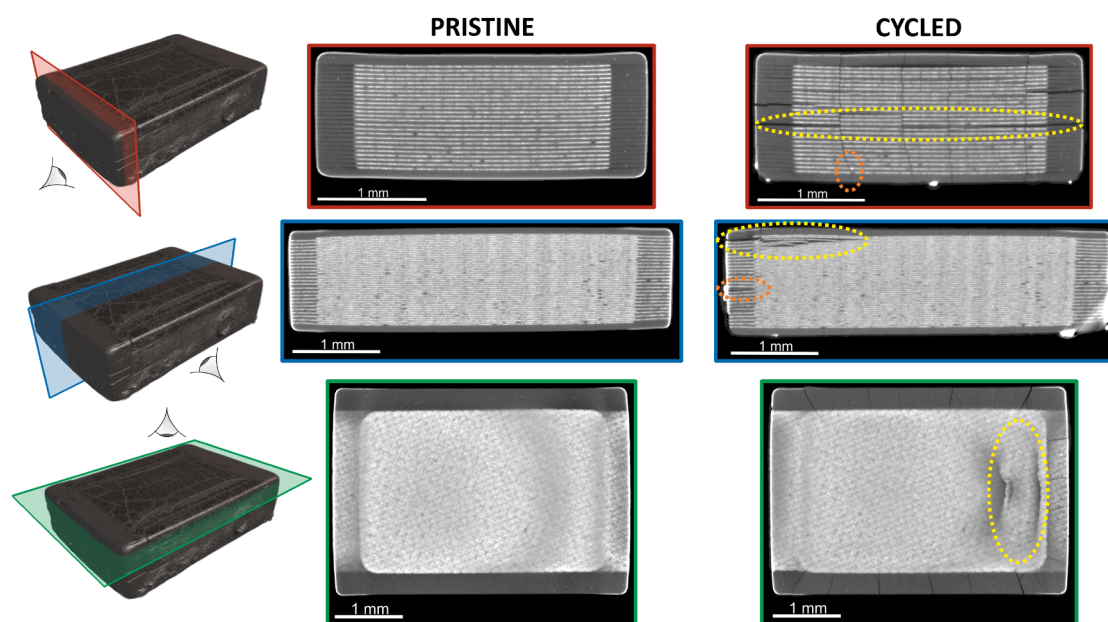


Figure 8. CT images of pristine and cycled cell-02

The cell was scanned at a fully discharged state at the corresponding slices with a $4.8\ \mu\text{m}$ voxel size. The cycle process was carried out under 0.1 C and 100 cycles. The red (XZ plane), blue (YZ plane), and green (XY plane) slices are displayed on the top, middle, and bottom regions.

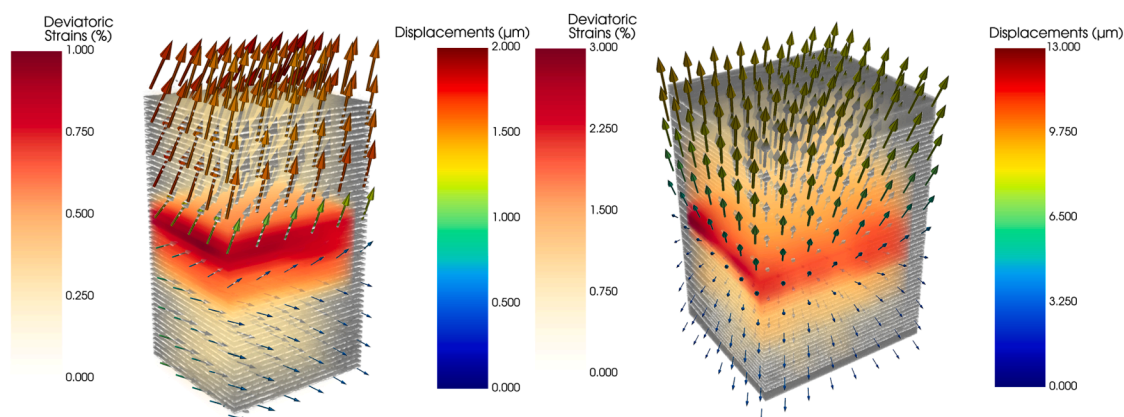


Figure 9. Result of DVC analysis that uses both the before and after cycling image dataset to reveal how much strain was induced by cycling Deviatoric strain results are shown in a white-red color map (left), and the displacement magnitude in blue-green-red (right) color maps of the quiver plot. Left: cell-01; right: cell-02.

delamination of the cathode layers and the SE from the current collector. From this *ex situ* investigation, a more comprehensive study could follow, i.e., an *in situ* CT study, to understand in more detail how the appearing damages occurred, which could then be related to the lifetime prediction of SSBs.

Quantitative evaluation

The availability of CT images from both pristine and damaged SSBs allows for a quantitative evaluation of their structural differences through DVC analysis.^{34,36} In essence, it works like the digital image correlation (DIC) technique, where a particular pattern is tracked through time in order to obtain the strain field in 2D, whereas DVC uses the image data described in each voxel to obtain the strain field in 3D. We used an open source module named SPAM^{34,35} to perform the DVC of cell-01 and cell-02. As depicted in Figures 6C and 6D, the input images of cell-01 are located around the center region of the cell and the edges region for cell-02. These regions were chosen with the hypothesis in mind that cell-02 should have a larger strain in comparison to cell-01 due to the significant delamination as shown in Figure 8. For DVC evaluations, the images taken with the LCT-02 parameter were used for cell-01 and those with the SCT parameter for cell-02. Due to computational limitations, the input images for cell-02 were down-sampled to the voxel size of 1.44 μm. Both sets of input images were normalized to 0 and 1, so that the peak original grayscale values coincide with one another. From a simple observation, a small difference in cracks and delamination in Figure S18A can be seen and a larger damaged region is even more pronounced in Figure S18B.

The DVC analysis started with the image registration process between the pristine and damaged images, which was carried out in two steps. First, a rough manual eye registration and linear-mutual information algorithms in Dragonfly were used to obtain the rough initial image registration. Afterward, the built-in registration function in SPAM computed the correlation function that describes the translation, rotation, and expansion between the two images. Then, the local DIC algorithm in SPAM was executed, where a regular-grid spacing must be inserted as an input. A node spacing of 30 pixels was used for the analysis for cell-01 and 40 pixels for cell-02. Roughly around 85% and 90%

of the computed voxels converged with the stopping criterion of 0.001, and the rest of them required some filtering in the correlation function to obtain smoother displacement fields. SPAM has the median filter function, where 27 pixels' radius was evaluated for cell-01 and 9 pixels' radius for cell-02. The isotropic and deviatoric strain invariants were calculated under the small strain theory framework, where the mean volumetric and deviatoric strain invariants can be expressed as follows:

$$\epsilon_{vol} = \text{tr}(\epsilon) = \frac{1}{3}(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}). \quad (\text{Equation 1})$$

$$\epsilon_{dev} = \|\epsilon - \epsilon_{vol}\|. \quad (\text{Equation 2})$$

Figure 9 displays the results of DVC analysis from both cell-01 and cell-02, which essentially showed a similar strain development. The strain originated around the middle region of the SSBs, where the average deviatoric strain for cell-01 was $0.21 \pm 0.18\%$ and for cell-02, $0.87 \pm 0.59\%$. This means that cell-02 has seen around 4 times higher strain than cell-01. Additionally, the quiver plot shows the displacement vector describing the local movement caused by the corresponding strains. The average displacement of cell-01 is $1.15 \pm 0.42 \mu\text{m}$, whereas that for cell-02 is $6.83 \pm 3.34 \mu\text{m}$, which is around 6 times higher than that for cell-01. These results imply that the larger mechanical damage could have led to delamination (depicted in Figure 8). In addition, the direction of the quiver in both cells showed a larger displacement to the upper side. Looking at the difference in the quiver size and color, cell-02 showed a higher displacement gradient to the upper side than cell-01 as observed in the CT images. The displacement in cell-02 looked like an explosion, i.e., equal in all directions, which originated around the center region, where its magnitude appeared to be increased gradually from the lower to the upper region, whereas in cell-01, the displacement appeared to have more variation in its directions and magnitudes.

Summary of ex situ investigation

Based on the CT scan results, we observe both volume expansion and fractures during the degradation of the SSB. From the cells' quantitative evaluation, the developed strain inside

cell-01 after 100 cycles was $0.21 \pm 0.18\%$ and $0.87 \pm 0.59\%$ for cell-02, with their respective displacement as $1.15 \pm 0.42 \mu\text{m}$ and $6.83 \pm 3.34 \mu\text{m}$. This volume expansion and mechanical contact loss among layers is one of the most detrimental mechanisms for oxide-based SSBs. Tippens et al.³⁷ reported the evolution of mechanical damage within lithium aluminium germanium phosphate SE caused by interphase growth during electrochemical cycling. The expansion of the interphase volume induces fractures, and the degree of fracture during cycling is identified as the primary contributor to the increase in impedance, rather than the resistance of the interphase itself. The area of the damaged SE grows proportionally to the value of the impedance. This finding provides strong support for the results from the electrochemical perspective for our ML model. Since mechanical degradation primarily drives the degradation, which is reflected in the growth of the impedance, i.e., the electrochemical data, the ML model using these data as inputs can precisely predict a cell state of health, based on the capacity trend in our case.

The degradation hinders the charge transfer reaction at the electrode-electrolyte interfaces, which results in significant impedance growths in the middle- to high-frequency ranges (15 Hz–1.5 kHz, i.e., in the 10^{-4} and 10^{-2} s time constant range in the DRT analysis). Besides, nonlinear mechanical failure can also happen during degradation. The qualitative CT evaluation demonstrates a localized delamination and fracture among layers in cell-02. This cell was not able to withstand mechanical strain beyond a certain point and partial failure at the intersection of electrode-SE and current collector occurred. This finding coincides with the sudden impedance increase observed in some EIS tests. Additionally, oxide-based SE can undergo decomposition when exposed to elevated electrode potentials and react with electrode materials at high potentials, resulting in the loss of active material, formation of undesirable interfaces, and the loss of lithium inventory. These coupled degradation processes also contribute to the overall impedance growth and capacity loss. Extending the present framework toward operando correlation between impedance features and mechanical degradation, as demonstrated in recent studies,³⁸ represents an important direction for future work.

Summary of findings

We present a comprehensive approach for battery capacity prediction by leveraging EIS, DRT analysis, and ML models for a commercial SSB, following a supplement by CT techniques. The data generation and preprocessing phase tackled the issue of data scarcity by ensuring high-quality and consistent impedance measurements across various cycling conditions. We processed a total of 112 datasets from eight batteries, utilizing the impedance data collected at frequencies higher than 0.2 Hz for feature engineering. The exclusion of low-frequency impedance data was therefore motivated by fundamental EIS reliability considerations rather than empirical feature selection, ensuring that the retained data satisfy the assumptions required for both DRT analysis and data-driven modeling. ML models, including BR, GPR, NN, and DT, were trained to diagnose the current capacity and predict the relative capacity after 10 more cycles. Using DRT for further feature engineering, we revealed that the impedance

has a more significant growth in the middle- to high-frequency range. When combined with DRT feature selection, some models demonstrated even better diagnostic accuracy with an MAE of 1.6% and robust performance in predicting future capacity with an MAE of 2.4%. Integrating DRT-based feature selection significantly accelerated BR and GPR model training and improved the physical interpretability of the aging mechanisms by focusing on relevant frequency ranges. The impedance data between 1.5 kHz and 15 Hz were found to be particularly informative for prediction performance in BR and GPR models. Finally, through non-destructive CT scans, we further link the observed impedance changes to physical degradation mechanisms, such as mechanical failures and volume expansion in the cells. While the present study is based on a limited number of cells, this reflects the high experimental cost and long time-scales associated with SSB aging experiments. Validation on larger and more diverse datasets will be an important next step to fully establish generalizability. This combined approach provides a methodological foundation that could be extended to industrial battery applications and quality control, as it not only helps improve the efficiency of battery testing—with high-frequency impedance tests taking only a few seconds—but also enhances the computational speed of battery prediction models by providing more refined and precise data related to cell degradation. The expected high-frequency impedance response can be separated almost effortlessly from the other signals and noises, which could be a promising method for battery diagnoses in various applications. All in all, this study highlights the effectiveness of combining impedance, DRT analysis, and ML for battery capacity prediction with very small datasets. The method developed is highly accurate and efficient and shows potential for scalability toward broader applications in other battery chemistries and electrochemical systems, pending further validation. The successful integration of data-driven approaches with electrochemical insights marks a significant step toward more reliable and faster battery diagnostics, with relevance to future industrial applications, such as battery management systems.

METHODS

Cell study and analysis

Eight commercial rechargeable SSB cells from TDK Electronics were cycled for the DRT study. Two of the eight cells were investigated by XCT. These two cells (cell-01 and cell-02) were cycled at the same rate of 0.1 C. Besides the eight cells, one cell was studied, only to design the test procedure, which is not included in the ML modeling and the DRT study. The cell dimensions are $4.4 \times 3.0 \times 1.1 \text{ mm}^3$; it is designed for applications that require lower power but compact, reliable, and extremely safe power supplies, e.g., for the industrial internet of things. In our setup, the cells were reflow soldered on a self-designed PCB board, as shown in Figure S1. Electrochemical cycling and EIS testing used a Biologic BCS-810 cycler in a constant temperature (25°C) and humidity (30%) environment. The cells were cycled at various C-rates, which were 0.1, 0.25, 0.5, and 1 C.

The manufacturer's recommended operating voltage range is 0–1.6 V, with a nominal capacity of 100 μAh and a nominal

discharge current of 20 μA . The discharging capability of these cells is up to 1000 μA . As there are only a few commercial SSBs on the market, which vary quite far from materials to cell concepts, there are no standardized categories to evaluate the performance of SSBs. The operational range and performance given by the cell manufacturer are designed for stable and long-life applications. For our purposes, it is more meaningful and efficient to test the cell with its full potential range, i.e., extended beyond the standard specifications, as the goal of this work is to develop a characterization and diagnosis method for all degradations expected to be seen during cycling. The manufacturer reported that the cell has a symmetrical structure and a Li-based ceramic oxide electrolyte.³⁹ As the cells, when received, are not formed and have no marked positive and negative electrodes, the material of cathode and anode should be the same, and the materials' distribution is assumed to be symmetrical as well, which was confirmed in our experiments. This symmetrical characteristic allows for cell operation down to 0 V.

Due to the low conductivity of the ceramic electrolyte, the cell was tested with a CCCV charge and CCCV discharge protocol to make sure the cell was fully discharged. In the potential window capability test (Figure S2), the cell has at least two plateaus in the charging progress, which is similar to phenomena in previous studies.^{18,19} These studies also pointed out that the degradation mechanisms are dependent on the reaction mechanisms, which are a combination of solid solution transition and two-phase behavior. During the cycling progress, the plateaus have drifted with cell aging (Figure S3), which could be related to the irreversible degradation in the material structure¹⁹—in line with the structural changes seen in our CT measurements. Due to the symmetrical cell concept and SE used, the cell has strong relaxation in certain potential windows. The EIS tests are also dependent on the cell's potential and state of charge. As Figure S4A shows, during the EIS test, the cell was not stable at a certain potential, especially under 1.5 V. The cell's impedance decreases with increased potential. In the high potential range, i.e., at 1.5 and 1.6 V, the linearity and stability of the EIS test got worse, as shown in Figure S4C. At the unstable states, the tested EIS results are difficult to evaluate, as reactions or diffusion were still going on during the EIS test. So, a certain relaxation before the EIS test is necessary to let the cell firstly "calm down." Previous studies also pointed out the relationship between impedance and relaxation time in SSBs⁴⁰ in a certain potential window. With this inspiration, the EIS tests were conducted with the cells charged to different capacities. The impedances, after different relaxation times, were compared in Figure S5. This study suggests that fully discharged cells exhibit the most stable impedance. Therefore, we only incorporated impedance data from fully discharged cells into our modeling and evaluation. Based on these studies, the eight cells were formed and cycled between 0 and 2.4 V with CCCV protocols, until their 100th cycle life, with EIS tests every 10 cycles in between. The tested EIS range is 10 kHz to 10 mHz. The rated capacity was the discharged capacity after formation, which is the 5th discharged capacity. The variable in the operation was the charging and discharging currents, from 20 to 200 μA . The cycling protocol is given in Figure S6.

DRTs

The DRT approach is a method used to extract additional information from EIS. In this approach, the distribution of relaxation times is calculated by transforming the impedance spectrum from the frequency domain to the time domain. This transformation provides increased resolution and more separated peaks. Subsequently, individual peaks can be extracted and used for analysis.

For the impedance, it holds:

$$Z(\omega) = R_0 + \int_0^\infty \frac{\gamma(\tau)}{(1+j\omega\tau)} d\tau$$

This equation can be simplified to:

$$Z(\omega) = R_0 + A(\omega, \tau) * \gamma(\tau)$$

The calculation of the DRT, $\gamma(\tau)$ proves to be an ill-posed problem here since $|A(\omega, \tau)| \approx 0$. Therefore, a regularization is necessary to obtain a suitable result. There are many different methods, which have been compared previously.³¹ In this work, the Gold algorithm is used. This algorithm is iterative and achieves regularization by reaching a stop criterion or an iteration maximum. The algorithm was chosen because it ensures robust and reproducible results without further conditions. Since only positive values are taken as input, the DRT spectrum also yields only positive values. The algorithm is given by:

$$\gamma_{k+1} = \gamma_k \frac{A^T A A^T \cdot Z}{\sum A^T A A^T A \cdot \gamma_k},$$

where k represents the iterations. Several iteration maximums were tried as regularization parameters, and one was eventually chosen, which showed good results in the application.

It should be mentioned that the DRT methodology is usually used to separate the spectra into many individual peaks. Here, however, the methodology is used as a key extraction method prior to the ML approach.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Daniel Schröder (d.schroeder@tu-braunschweig.de).

Materials availability

The tested batteries are commercial batteries. This study did not generate new materials.

Data and code availability

The battery cycling data are available on the Zenodo platform: <https://doi.org/10.5281/zenodo.16313223>. The code for data processing and modeling in this study is available at: <https://doi.org/10.5281/zenodo.18925051>. All other information is available from the lead contact upon request.

ACKNOWLEDGMENTS

We gratefully acknowledge the project DALION 4.0 (03ETE017A) and the Federal Ministry for Economic Affairs and Energy for their financial support for parts of this work. We thank Dr. Katja Kretschmer for her valuable advice on the conception of this research and for coordinating the preparation of the test materials. We would like to thank TDK Corporation for valuable

discussions regarding the commercial battery, particularly Yongli Wang, for his helpful suggestions on testing. We want to thank Dr. Tobias Art for his fruitful insights on the tomography measurement. We also appreciate the insightful discussions with Dr. Nicolas Schlüter on EIS measurements and DRT analysis.

AUTHOR CONTRIBUTIONS

Conceptualization, methodology, battery cycling and EIS experiments, data curation, formal analysis, modeling, visualization, writing – original draft, and writing – review and editing, B.Z.; X-ray experiments, image analysis and visualization, formal analysis, writing – review and editing, M.P.W.; DRT analysis method and writing – review and editing, T.B.; data preprocessing, Y.Y. X-ray experiments, M.E.; conceptualization, resources, supervision, and review and editing, E.J.; conceptualization, methodology, funding acquisition, project administration, resources, supervision, and writing – review and editing, D.S.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.xcrp.2026.103334>.

Received: December 1, 2025

Revised: March 23, 2026

Accepted: May 4, 2026

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